

Section 1.2:

Components of a Computer System and Storage Devices

In this section we will look at the different components (parts) of a computer system.

You will see what role each component plays within the computer system. You will also get to understand the features of each component.

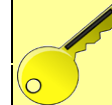
This section also briefly introduces the different forms of storage devices that we use to hold out data.

Learning objectives of this section:

- # Know what a **computer system** is and be able to **identify and describe** the main components of a computer system.
- # Understand what **internal memory** is and be able to **describe the difference** between RAM and ROM.
- # Be able to **identify and describe the difference between** input devices, storage devices and backing storage.

Computer Components

Definition: *"All the different pieces of electrical **hardware** that join together to make up the complete computer system."*

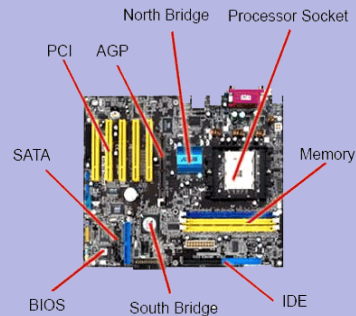


Key Words:

Motherboard, CPU, RAM, ROM, BIOS, BUS Network, Hertz, Graphics Card, Peripheral, Volatile, Non-Volatile, Storage Device

Examples of common Computer Hardware (components): (click images to zoom and retrieve more info)

Motherboard



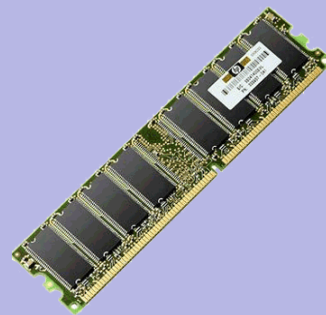
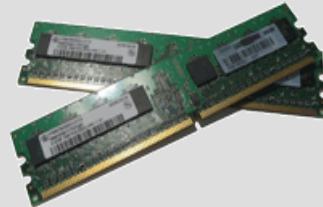
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CPU



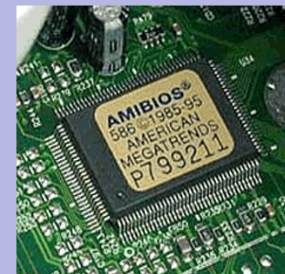
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RAM



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BIOS (ROM)



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Graphics Card



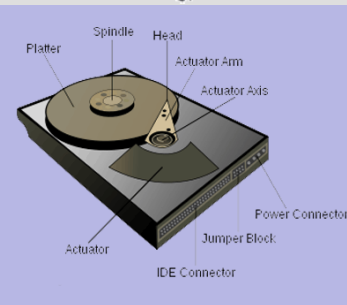
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Peripherals



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Storage Devices



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What is a computer system?

Components form the complete computer system. A computer system is made up of 4 main types of components:

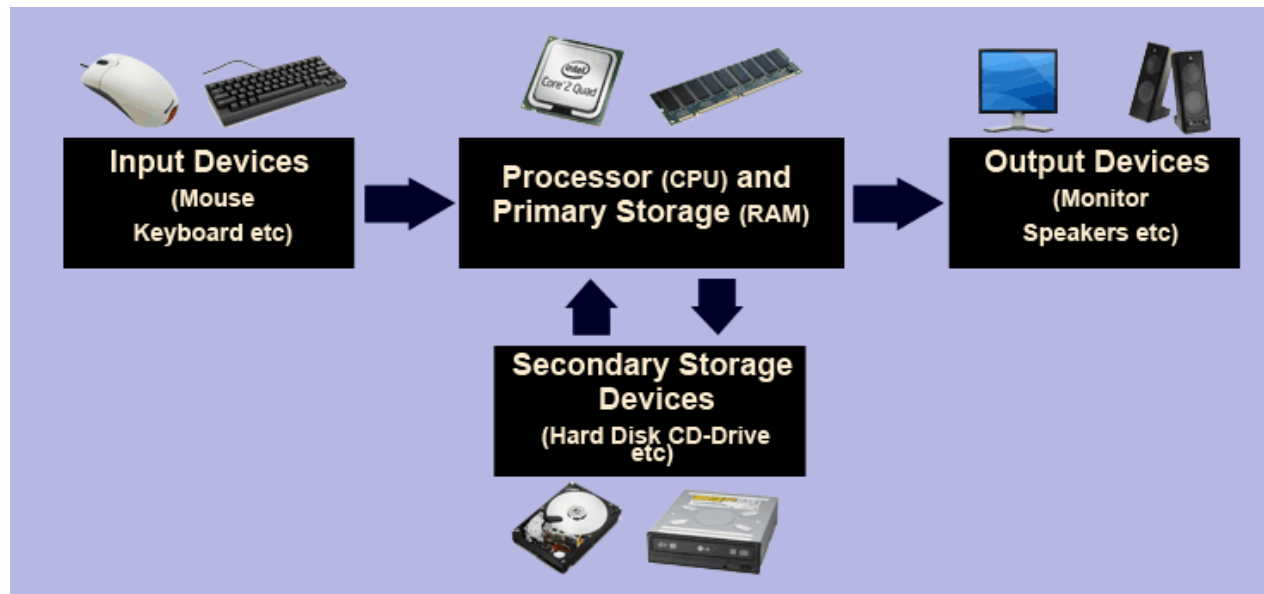
- Input Devices (keyboard, mouse etc)
- Output Devices (monitor, speakers etc)
- Secondary Storage Devices (hard disk drive, CD/DVD drive etc)
- Processor and Primary Storage Devices (cpu, RAM)

NOTE:

We will look at storage devices in greater detail in a later unit.

[Click for Example](#)

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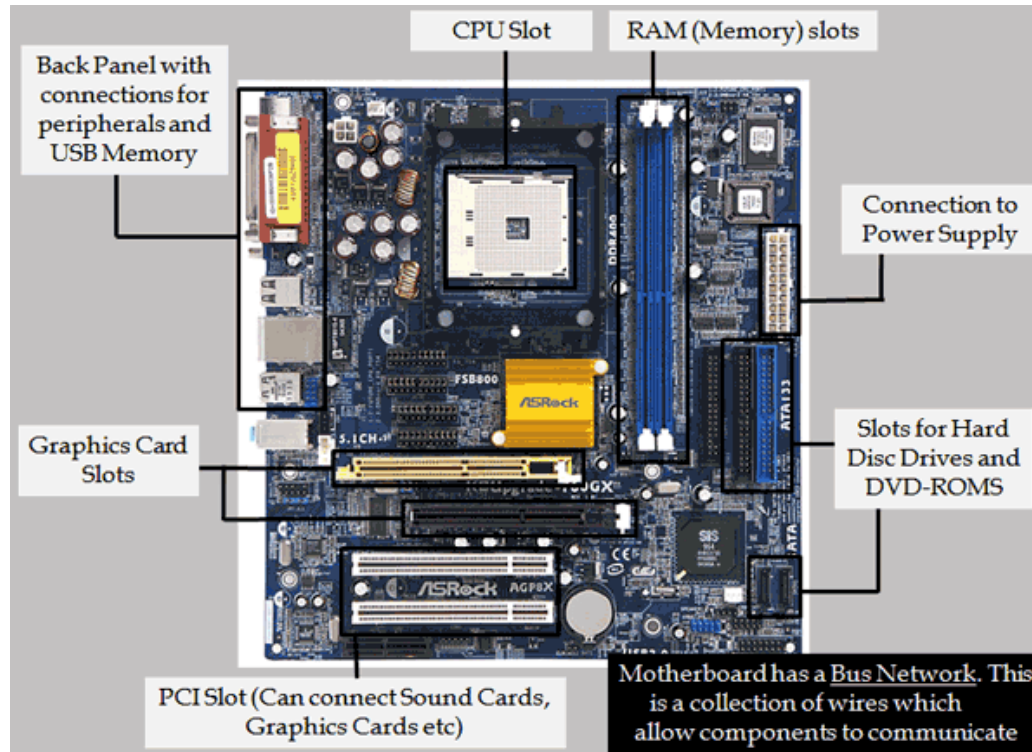
Features of Internal Hardware Computer Components

Internal computer components are designed to fit **INSIDE** the computer system and they all carry out important roles. We will discuss the following:

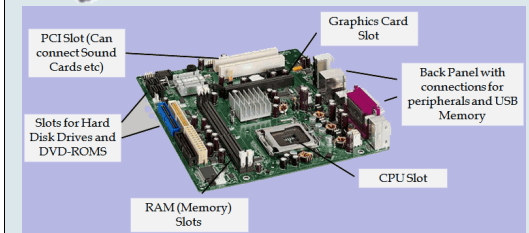
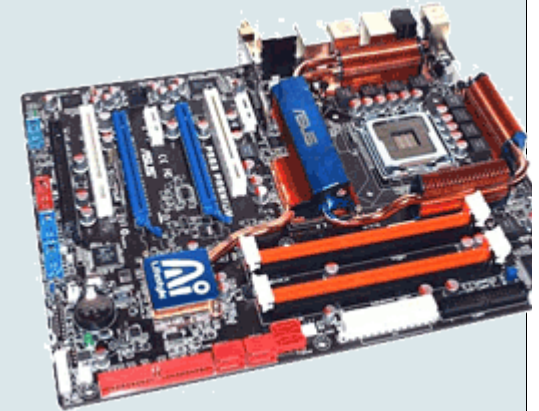
- **Motherboard** (this does something with the data to make it useful information)
- **Processor** (central processing unit)
- **Internal Memory** (RAM and ROM)
- **Video Card** (aka graphics card)
- **Sound Card**
- **Internal Hard Disk Drive**

1 Motherboard

- # The motherboard is **central** to any computer system.
- # All components **plug into the motherboard** either directly (straight into the circuit board) or indirectly (via USB ports).
- # Once connected to the motherboard, the components can **work together** to form the **computer system**.
- # Components communicate and **send signals** to each other via the **BUS Network**.



Examples:

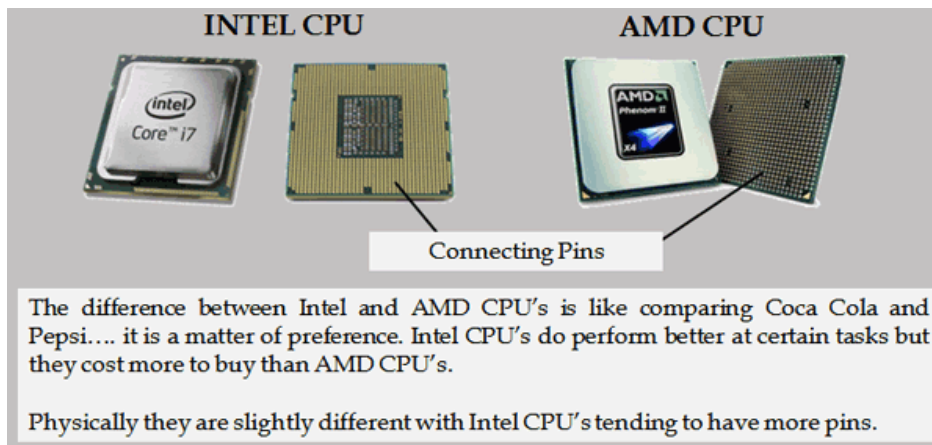


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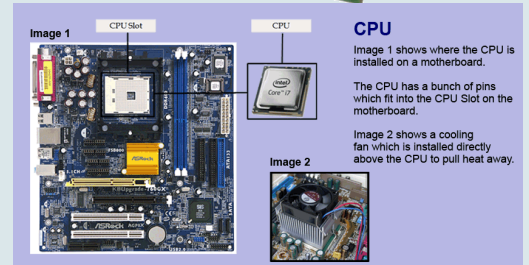
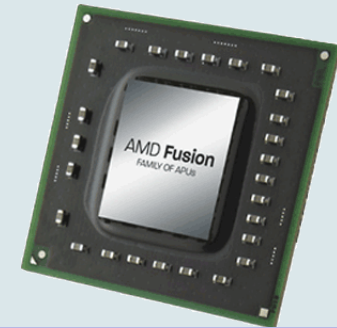
Click image to zoom in and retrieve more information.

2 Processor (CPU / Central Processing Unit)

- # The Central Processing Unit (CPU) is the **brain of the computer**.
- # The CPU '**controls**' what the computer does and is responsible for performing **calculations** and **data processing**. It also handles the movement of data to and from system memory.
- # CPU's come in a variety of speeds which are known as '**clock rates**'. Clock rates are measured in '**Hertz**'. Generally, the faster the clock rate, the faster the performance of the computer.
- # There are two main brands of CPU currently on the market... AMD and Intel:



Examples:



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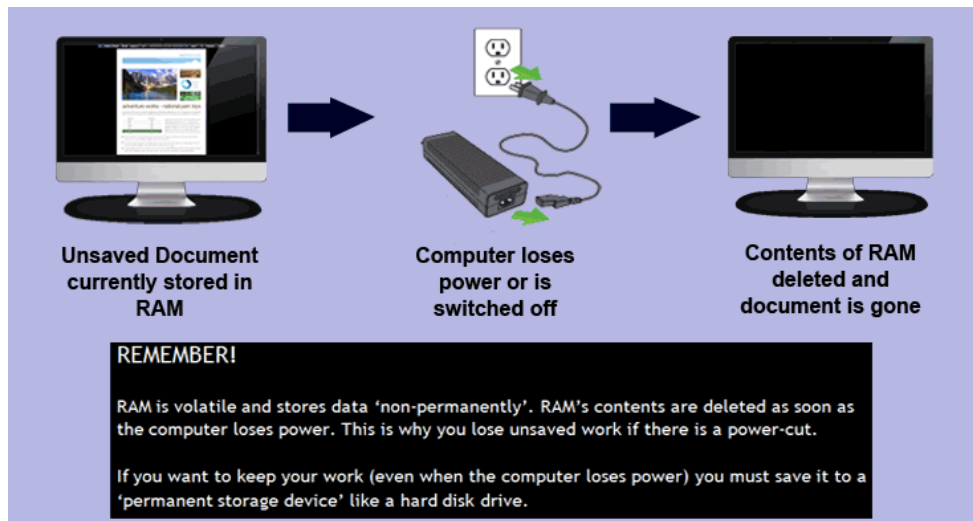
Click image to zoom in and retrieve more information.

3 Internal Memory (RAM and ROM)

- # There are two types of internal memory - **RAM** and **ROM**.
 - # RAM and ROM are used to **store computer data** and this can be directly accessed by the CPU.
 - # RAM and ROM are sometimes referred to as '**Primary Storage**'.
-

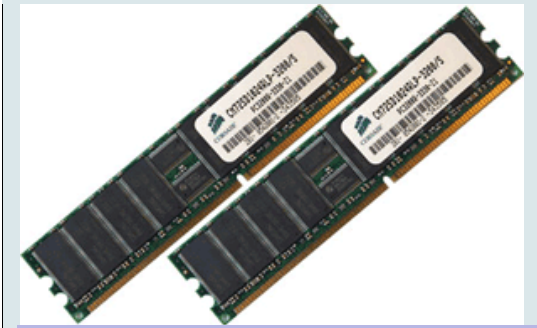
RAM (Random Access Memory)

- RAM is used to **temporarily store information** that is **currently in use** by the computer. This can include anything from word documents to videos.
- RAM can be **read from and written to** and so the information stored in RAM can change all the time (it depends what tasks you are using the computer for).
- RAM is a **fast memory**. Data can be written to and read from RAM very quickly. RAM is generally measured in GB (Gigabytes).
- RAM is **Volatile Memory**. This means that information stored in RAM is deleted as soon as the computer is turned off.



- The **more RAM** you have installed in your computer -- the **faster it can perform**. You can open and use more programs at the same time without slowing the computer down.

Examples:



REMEMBER!

RAM is volatile and stores data 'non-permanently'. RAM's contents are deleted as soon as the computer loses power. This is why you lose unsaved work if there is a power-cut.

If you want to keep your work (even when the computer loses power) you must save it to a 'permanent storage device' like a hard disk drive.

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Click image to zoom in and retrieve more information.

NOTE:

se up all of your RAM by opening too many programs and you will notice that your computer becomes very slow.

ROM (Read Only Memory)

- ROM is used to **permanently store instructions** that tell the computer how to **boot** (start up). It also **loads the operating system** (e.g. Windows).

These instructions are known as the **BIOS** (Basic input/output system) or the **boot program**.

- Information stored in ROM is known as **READ ONLY**. This means that the contents of ROM cannot be altered or added to by the user.
- ROM is **fast memory**. Data stored in ROM can be accessed and read very quickly.
- ROM is **Non-Volatile** memory. This means that stored information is not lost when the computer loses power.
- Other examples of ROM include:
 - **DVD/CD ROMS** bought in stores containing pre-recorded music and movie files. These are played back at home but cannot be altered.
 - **ROM** in printers which is used to store different font types.

Examples:



Examples of ROM



BIOS Chip with instructions which boot computer and load operating system.



DVD / CD ROMs containing pre-loaded music and movie content



ROM Chip in printers which store pre-loaded font styles.

Remember!

ROM is read only and cannot be altered by the user. This prevents accidental deletion of important commands which devices need to operate correctly.

**READ
ONLY
MEMORY**

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Click image to zoom in and retrieve more information.

NOTE:

The computer would not start-up without ROM Bios.



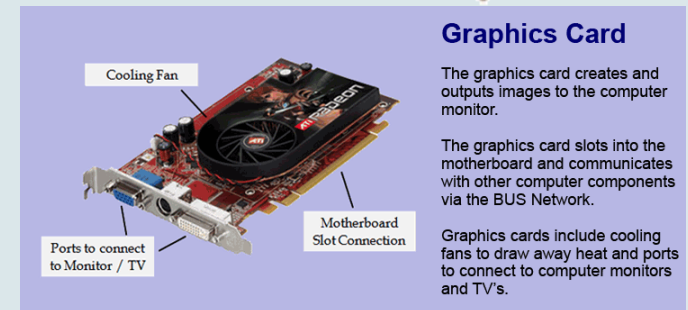
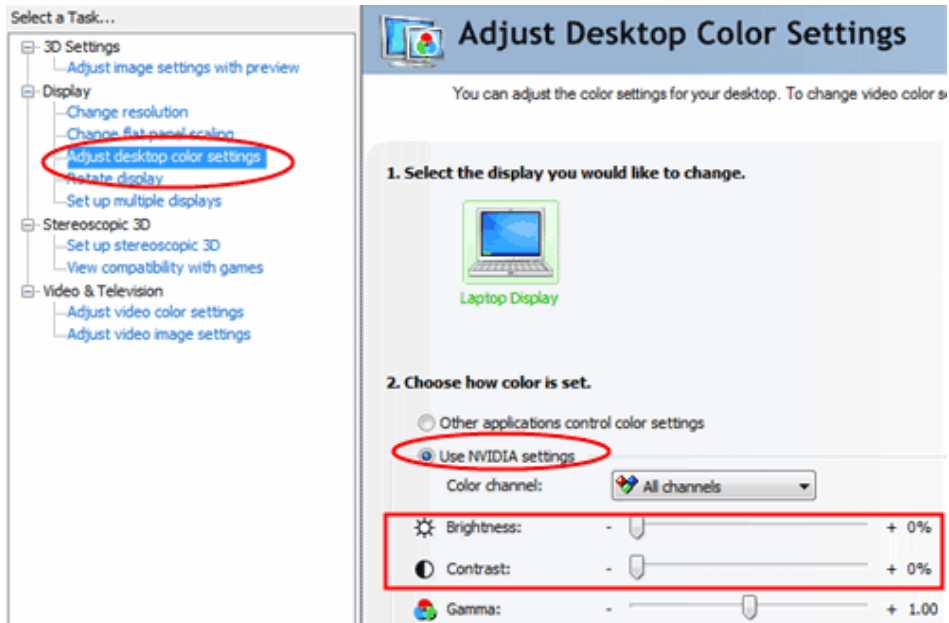
How to remember the difference between RAM and ROM.

		RAM	ROM
1.	Contents are lost when the computer is turned off.	✓	
2.	Contents are not lost when the computer is turned off.		✓
3.	Stores instructions that tell the computer how to start up.		✓
4.	Stores data and programs currently in use.	✓	
5.	Your computer will perform faster with more of this.	✓	
6.	This cannot be written to (altered).. only read.		✓
7.	Stands for Random Access Memory.	✓	
8.	Stands for Read Only Memory.		✓

4 Video Card (aka graphics card)

- # Graphics cards are hardware devices that plug into the motherboard and enables the computer to **display images on the monitor**.
- # Graphics cards usually require the **installation of software** alongside the hardware. The software **instructs the computer how to use the graphics card** and also allows you to **alter settings** to change image quality and size.
- # See below for an example of graphics card software allowing the user to alter various graphical settings:

Examples:



Graphics Card

The graphics card creates and outputs images to the computer monitor.

The graphics card slots into the motherboard and communicates with other computer components via the BUS Network.

Graphics cards include cooling fans to draw away heat and ports to connect to computer monitors and TV's.

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5 Sound Card

Sound cards are **internal hardware devices** that plug into the **motherboard**

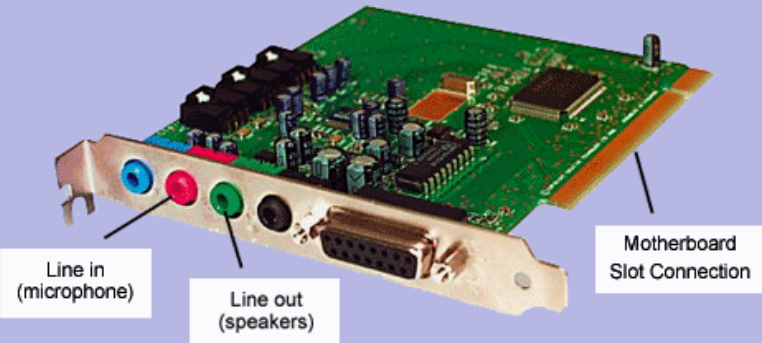
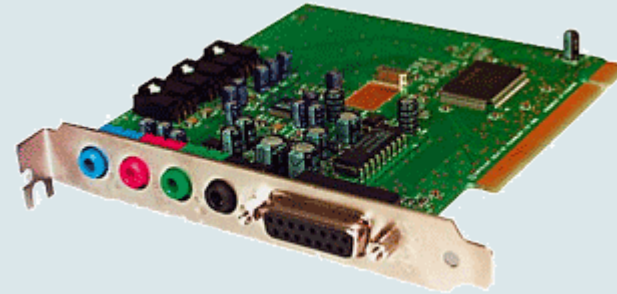
A sound card's main function is to allow the computer system to **produce sound** but they also allow users

Examples:

to **connect microphones** in order to **input sounds** into the computer.

- # Sound cards are also useful in the **conversion of analogue data into digital** and **vice versa**.

This topic is discussed in more detail in later units.



Sound Card

The sound card is used to produce digital sound and output it through external speakers.

Microphones can also be plugged in to input sound into a computer.

The sound card is important in the conversion of digital and analogue data.

The sound card is an **INTERNAL HARDWARE DEVICE**.

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6 Storage Devices (secondary backing storage)

- # Secondary storage devices are used to store data that is **not instantly needed** by the **computer**.
- # Secondary storage devices **permanently store data and programs** for as long as we need. These devices are also used to **back-up data** in case original copies are lost or damaged.

Examples:

Remember:

Temporary storage like **RAM** does not hold data for long periods.

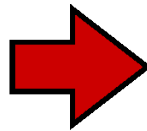
It is used to store only those programs and files that we are **currently working on**.

There are two categories of storage devices:

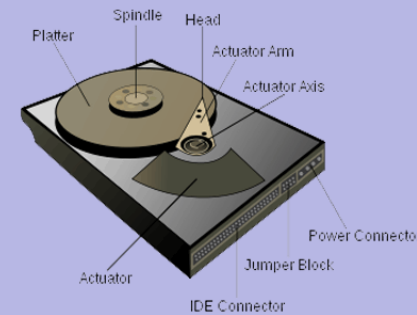
- **Internal Storage** - Internal Hard Disk Drives
- **External Storage** - External Hard Disk Drive, Memory Stick etc

Example of how a hard disk drive works:

The video to the right shows how data is read/written to the hard disk drive by spinning a platter across a read/write head.



The read/write head is on the end of a movable arm.



Hard Disk

Hard Disks are usually fixed into a computer and are used to permanently store data.

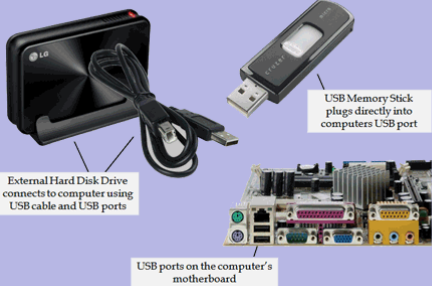
Hard Disks vary in capacity but newer versions allow for huge amounts of storage.

Hard Disks are known as **PERMANENT STORAGE DEVICES**.

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Click image to zoom in and retrieve more information about **INTERNAL STORAGE**.



External Hard Disk Drive connects to computer using USB cable and USB ports

USB Memory Stick plugs directly into computers USB port

USB ports on the computer's motherboard

External Storage

External storage devices perform the same function as internal storage devices --- to **permanently store data**.

The only real difference is that external storage devices are **easy to disconnect from the computer** in order to be moved and used somewhere else.

External storage devices include **External Hard Disks and Memory Sticks**.

External storage devices normally use **USB** to easily connect / disconnect from the computer.

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Click image to zoom in and retrieve more information about EXTERNAL STORAGE.

Features of External Hardware Computer Components

External computer components connect to a computer system from **OUTSIDE**. They are not necessary for the system to function but make our experiences easier or better. We will discuss the following:

- **Input Devices** (used to get data into a computer)
- **Output Devices** (used to get information out of a computer)
- **Peripherals**

1 Input Devices

- # Input devices are pieces of hardware that get **raw data into the computer** ready for processing.
- # Processing involves taking raw data and turning it into more useful information.
- # Input devices fall into two categories:
 - **Manual Input Devices** - Need to be operated by a human to input information
 - **Automatic Input Devices** - Can input information on their own.

Manual Input Devices

Keyboard



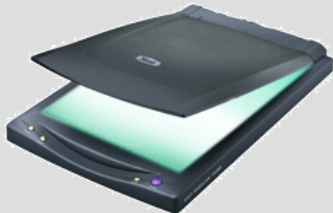
Mouse



Microphone



Scanner



Numeric Keypad



Light Pen



Touch screen



Digital Camera



Web Cam



Automatic Input Devices

Magnetic Stripe Reader



Magnetic Ink Character Recognition



Chip and Pin Reader



Optical Mark Reader



Optical Character Reader



Barcode Reader



NOTE:

We will look at input devices in more detail during a later unit!

Click [here](#) to jump to the unit

2 Output Devices

- # When inputted raw data has been processed it becomes **usable information**. Output devices are pieces of hardware that send this usable information **out of the computer**.
- # Some output devices send information out temporarily and some send information out permanently:
 - **Temporary Output Devices** - E.g. Monitors which constantly refresh the outputted image on the screen
 - **Permanent Output Devices** - E.g. Printers which output information onto paper as a hard copy.

Output Devices

CRT Monitor



TFT Monitor



Laser Printer



Inkjet Printer



Dot Matrix Printer



Plotter



Speakers



Multimedia Projector



Note:

We will look at output devices in more detail during a later unit!

3 Peripheral Devices

- # Almost all input and output devices are known as '**Peripheral devices**'.

- # These are '**non-essential**' hardware components that usually connect to the system **externally**.
- # Peripherals are called non-essential because **the system can operate without them**.

For example : You can still use your computer without speakers... you will just have to do without high quality sound.

- # Examples of peripherals are shown below:

Peripheral Devices

Keyboard



Mouse



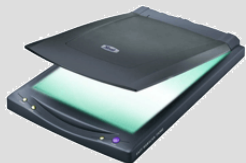
Microphone



Web Cam



Scanner



Joystick



Speakers



Printer



Examples:



Peripherals

Peripherals are 'non-essential' pieces of hardware that plug into the computer externally.

They are called 'non-essential' because a computer can operate without them.

Examples of peripherals include speakers, microphone, mouse, joystick, printer, web-cam etc.

Almost all peripherals connect to the computer externally via USB.

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Click image to zoom in and retrieve more information.

Activity!

Task 1 Computer Components

Click the above task and complete the diagram of a computer system.

Activity!

Task 2 Identifying Components

Click the above task and follow the instructions given on the sheet.

Activity!

Task 3 Motherboard and Components

Click the above task and arrange the components into the correct place on the motherboard.